Day-Ahead Flight Arrival-Delay Risk for NYC Departures: Leakage-Aware Features, Logistic Regression versus Gradient Boosting, and Calibration under Shift

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***Abstract*—We rank flights leaving New York (EWR, JFK, LGA; 2013) by the risk of arriving at least 15 minutes late, using only information known at 20:00 on the evening before departure. Two models (regularised logistic regression and XGBoost) are compared with two references (base-rate prior, route history). On the held-out Nov–Dec window the final model reaches PR-AUC 0.394 against 0.353 for route history and 0.258 for the prior. Flagging the riskiest 10% of each day gives precision 0.444 and lift 1.72. The two learned models are not statistically distinguished. A calibration rule fixed in advance fails in December (ECE 0.142), whereas a causal rolling recalibration tracks the seasonal shift. We separate selection evidence from exploratory test findings and state what a daily review-desk analysis does and does not show.**

***Index Terms*—flight delay prediction, data leakage, gradient boosting, probability calibration, temporal validation, decision support**

KEY FINDINGS

1) **Leakage is measurable.** 31.1% of the previous day's arrivals are unknown at 20:00; letting them in inflates CV PR-AUC by ≈0.011.
2) **Both models beat both references.** Test PR-AUC: LogReg 0.394, XGBoost 0.407, route history 0.353, prior 0.258.
3) **LogReg ≈ XGBoost.** CV 0.4705 vs 0.4450; test-difference CI $[-0.010, +0.041]$ includes zero. The final model is chosen by a rule declared before test.
4) **Static calibration breaks under season shift.** Test ECE 0.082 (static Platt) vs 0.027 (rolling Platt); December ECE 0.142.
5) **A capacity-limited desk gains from ranking.** The top 10% per day is 1.72× more precise than random (Sec. X); this is ranking quality, not realised savings.

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I. Introduction

Flight delays cascade: a late inbound aircraft, a congested departure bank or an upstream weather system turns one disruption into many. Airlines, airports and passengers all benefit from knowing *in advance* which departures are most at risk, because the remedies (adding buffer to a rotation, swapping an aircraft, re-timing a departure, pre-positioning staff) take time to execute. This report studies that day-ahead ranking problem on the public `nycflights13` data, with the emphasis on *doing it honestly*: every feature must be computable at the moment the prediction is issued, every choice must be justified without looking at the test period, and every probability must be checked for calibration.

A. Problem and motivation

- **Task.** For each flight scheduled to depart EWR, JFK or LGA on day D, estimate $P(\text{arrival delay} \geq 15 \text{ min})$ at **20:00 NYC time on** $D-1$, and rank the flights by that estimate.
- **Why a decision clock matters.** Every past outcome is present in the table, so a careless join, rolling window or lag silently lets the future into the features. A high score obtained that way answers a question no operator can ask in real life. The 20:00 clock fixes exactly which outcomes are usable.
- **Why probabilities and not only ranks.** A desk that triages by rank needs only the ordering, but a desk that sets a threshold, budgets buffer time or compares airlines needs *calibrated* probabilities. We therefore evaluate both ranking (PR-AUC, ROC-AUC, lift) and probability quality (Brier, ECE).
- **Why this data.** `nycflights13` covers one full year at three airports, which gives strong seasonality (calm autumn, extreme December). That makes a chronological split a built-in stress test of calibration under shift.

B. Contributions

- A **decision clock**: an outcome enters a feature only once it is known; the resulting leak is quantified and unit-tested (Sec. IV).
- A **controlled model comparison** of logistic regression and XGBoost [1] against two references, with the final model fixed by a rule declared before test.
- A study of **calibration under seasonal shift** [2], [3]: a pre-declared static rule and a causal rolling alternative.
- A **policy-level decision-support analysis** for a capacity-limited review desk (Sec. X).
- A **development protocol** that discloses which test numbers were seen, and when, and keeps selection evidence separate from exploratory findings.

C. Scope

- **In scope:** machine learning only; timetable features and outcomes already known at the decision time.
- **Out of scope:** weather, deep temporal models, and any cost or schedule-change model (Sec. XI-D). Weather is excluded *by design*: observed weather for D is not known at 20:00 on $D-1$.

TABLE I
Notation.

Symbol	Meaning
D	target (departure) day
$a(t)$	availability date of an outcome known at time t
$y \in \{0,1\}$	late-arrival label (arr_delay ≥ 15 min)
$m,\ w$	shrinkage strength, history window (days)
C	inverse L2 strength of logistic regression
$\hat{r}_{k,D}$	shrunk delay rate of entity k for day D
$g_i,\ h_i$	gradient and Hessian of the loss for sample i
$k\%$	daily review capacity (share of flights flagged)

D. Organisation

Section II reviews related work and states the research gap. Sec. III–V describe the data, decision clock and features. Sec. VI gives the models and calibration theory, Sec. VII the evaluation protocol. Sec. VIII reports the evidence behind every decision and Sec. IX the exploratory test results. Sec. X is the decision-support analysis, Sec. XI discusses limits and validity, and Sec. XII concludes.

E. Notation

Table I lists the symbols used throughout.

II. Related Work

This section places the work in six strands and ends with the research gap. Each paper cited here was confirmed to exist by a bibliographic search; items we could not fully confirm are marked VERIFY in the matrix and in `references.bib`.

A. Flight-delay prediction

- **Surveys.** Sternberg et al. [4] organise the literature by scope, data and computational method, and highlight the growing use of machine learning. This is the map against which our design (single airport group, one year, tabular model) can be placed.
- **Classical machine learning with explicit horizons.** Rebollo and Balakrishnan [5] forecast departure delays 2–24 hours ahead with random forests, combining *local* delay states with *network* delay variables describing the whole airspace. Two ideas carry over directly to our work: (i) the horizon is part of the problem definition, and (ii) recent network-wide state is a strong predictor, which motivates our day-level history group. Their target is delay minutes (regression) and the model reports median error; we instead rank flights and study calibration.
- **Scalable data mining and deep learning.** Belcastro et al. [6] apply scalable data-mining pipelines to flight delays, and Yu et al. [7] apply deep models to commercial-flight delays. We did not read the full texts of these two works, so we do not characterise their features or metrics in the matrix (marked VERIFY).

B. Gradient boosting and tabular data

- **Boosting.** Friedman [8] casts boosting as steepest descent in function space; XGBoost [1] adds second-order optimisation, explicit regularisation and a fast tree learner. We use it as the flexible learner against logistic regression.
- **Trees versus deep nets on tabular data.** Grinsztajn et al. [9] find that tree ensembles remain state of the art on medium-sized tabular data. That supports XGBoost as a serious Phase 1 baseline before any deep model.

C. Leakage

- **Formulation.** Kaufman et al. [10] define leakage as the introduction of information about the target that should not legitimately be available at prediction time, and discuss avoiding it through a clear *learn–predict separation*. Our decision clock is an operational version of that separation for a daily forecasting problem: availability dates enforce which outcomes may feed which prediction.

D. Probability calibration and shift

- **Post-hoc calibrators.** Platt scaling [11] fits a sigmoid to classifier scores; isotonic-regression-based calibration [12], [13] fits a monotone step function and is more flexible but more data-hungry; beta calibration [14] generalises the sigmoid with a richer map family. Guo et al. [2] popularise ECE and temperature scaling for neural networks.
- **Calibration under shift.** Ovadia et al. [3] show that uncertainty and calibration degrade under dataset shift; Rabanser et al. [15] study detecting shift (including domain classifiers, which is the idea behind our adversarial validation); Gama et al. [16] survey adapting to concept drift. These studies mostly use image or text benchmarks, not event probabilities with a seasonal base-rate change.
- **Scoring rules.** The Brier score [17] and log-loss are strictly proper scoring rules [18]: they are optimised only by the true probability, which is why we optimise log-loss and report Brier.

E. Imbalanced evaluation

- **PR versus ROC.** Davis and Goadrich [19] relate PR and ROC curves and show that optimising ROC area does not optimise PR area; Saito and Rehmsmeier [20] show PR plots are more informative on imbalanced data. Our base rate is about one quarter late, so PR-AUC is the primary ranking metric and ROC-AUC is reported alongside.

F. Temporal validation and uncertainty

- **Rolling origin.** Tashman [21] reviews out-of-sample tests and the advantage of rolling-origin evaluation over a single origin; Bergmeir and Benítez [22] discuss cross-validation for time series and suggest blocked forms. We use rolling-origin CV inside the training window.
- **Block bootstrap.** Künsch [23] introduces resampling *blocks* for dependent observations. Flights on one day are correlated, so we resample whole days (a block of one day).

G. Comparison matrix

Tables II, III, IV and V compare the works above in four tables (two column groups for each of two paper groups). "–" means not applicable; "not assessed" means the item was not read in the source, and VERIFY marks entries awaiting confirmation. The matrix records only what we could confirm.

H. Research gap

- **Gap 1 — an explicit decision clock.** Among the works above, none pairs a day-ahead decision time with an audit of which outcomes are known at that time; leakage is treated in general terms [10]. We make the clock explicit, quantify the leak and test for it.
- **Gap 2 — calibration under a seasonal base-rate change.** Calibration under shift is studied for deep networks on benchmark shifts [3]; we study delay probabilities across a real seasonal change and compare a pre-declared static calibrator with a causal rolling one.
- **Gap 3 — policy-level evaluation.** Delay studies mostly report aggregate scores; we evaluate a capacity-limited daily top-$k\%$ policy and say what it does not show.

These statements are limited to the papers we could confirm; we make no claim about the literature as a whole.

III. Data and Exploratory Analysis

We first audit the data, fix the target, and then read the exploratory patterns as design constraints. Table VI at the end of the section maps each observation to a design decision.

A. Data quality

- **Source.** `nycflights13` [24]: 336,776 departures from EWR, JFK and LGA in 2013, with carrier, flight number, tail number, origin, destination, distance, scheduled and actual times and delays.
- **Integrity checks (all passed).** No duplicate flights by identifier or by (date, carrier, flight, origin, scheduled time); the departure delay equals the clock-time difference modulo 24 h for every flight; hour and minute match the scheduled departure for every flight; no non-positive air time.
- **One anomaly.** EWR–EGE and JFK–EGE each have two distances (1726/1725 and 1747/1746) that change by exactly one mile on 1 March 2013: a dated reference-data update with no practical effect, so we keep the values.
- **Missing values are structural.** 8,255 flights were cancelled (no departure time), 1,175 diverted (departed, no arrival delay), and all 2,512 missing tail numbers belong to cancelled flights. These columns encode the outcomes we predict, so imputing them would be wrong and we **do not impute**.

B. Target

- **Label.** $y = \mathbb{1}[\text{arr_delay} \geq 15]$, the BTS definition of a late arrival. The positive rate is 24.5% of completed flights (16.1% at 30 min, 8.7% at 60 min).
- **Why arrival, not departure delay.** The arrival delay is what passengers and downstream connections experience;

TABLE II
RELATED WORK, FLIGHT-DELAY AND BOOSTING PAPERS (1/2): DATA, HORIZON, FEATURES AND MODEL.

Paper	Year	Dataset	Horizon / decision time	Weather features	Model
Rebollo & Balakrishnan [5]	2014	US air-traffic delay data	2–24 h ahead	not assessed	Random forest (regression)
Belcastro et al. [6]	2016	Flight records	not assessed	VERIFY	Scalable data-mining classifiers
Yu et al. [7]	2019	Commercial flights	not assessed	VERIFY	Deep learning
Sternberg et al. [4]	2017	– (review)	–	–	Taxonomy of methods
Friedman [8]	2001	– (method)	–	–	Gradient boosting machine
Chen & Guestrin [1]	2016	Several benchmarks	–	–	Gradient-boosted trees (XGBoost)
Grinsztajn et al. [9]	2022	Tabular benchmark (∼45 sets)	–	–	Trees vs deep nets
This work	2026	nycflights13 (2013)	20:00 on $D-1$	None (by design)	LogReg, XGBoost

TABLE III
RELATED WORK, FLIGHT-DELAY AND BOOSTING PAPERS (2/2): METRICS, CALIBRATION, TEMPORAL HANDLING AND LIMITATION.

Paper	Metrics	Calibration reported	Leakage / temporal handling	Limitation (for our purpose)
Rebollo & Balakrishnan	Median test error 21 min at 2 h horizon	– (regression)	Forecast horizon explicit	Minutes, not a ranked decision; no calibration
Belcastro et al.	not assessed	not assessed	not assessed	VERIFY
Yu et al.	not assessed	not assessed	not assessed	VERIFY (deep model, out of our scope)
Sternberg et al.	–	–	–	Survey; no new model
Friedman	Loss-specific fit	No	–	Not calibrated by design
Chen & Guestrin	Benchmark scores	No	–	Probabilities not guaranteed calibrated
Grinsztajn et al.	Benchmark accuracy	No	–	Not flight data
This work	PR-AUC, ROC-AUC, Brier, ECE, lift	Yes, static and rolling	Decision clock + leakage tests	Single year, no weather

TABLE IV
RELATED WORK, CALIBRATION, SHIFT, EVALUATION AND LEAKAGE PAPERS (1/2): DATA, HORIZON AND METHOD.

Paper	Year	Dataset	Horizon / decision time	Method
Brier [17]	1950	Weather forecasts	–	Brier score
Platt [11]	1999	SVM outputs	–	Sigmoid calibration
Zadrozny & Elkan [12]	2002	Benchmark classifiers	–	Calibration of classifier scores
Niculescu-Mizil & Caruana [13]	2005	UCI-style benchmarks	–	Many learners + Platt / isotonic
Kull et al. [14]	2017	Benchmark classifiers	–	Beta calibration
Guo et al. [2]	2017	Image / text benchmarks	–	Neural nets + temperature scaling
Ovadia et al. [3]	2019	Shifted benchmarks	–	Deep nets, uncertainty methods
Rabanser et al. [15]	2019	Benchmark shifts	–	Shift detection methods
Gama et al. [16]	2014	– (survey)	–	Drift adaptation
Gneiting & Raftery [18]	2007	– (theory)	–	Proper scoring rules
Davis & Goadrich [19]	2006	–	–	PR / ROC theory
Saito & Rehmsmeier [20]	2015	Simulated and real	–	PR vs ROC study
Tashman [21]	2000	– (review)	–	Out-of-sample tests
Bergmeir & Benítez [22]	2012	Time series	–	Blocked cross-validation
Künsch [23]	1989	Stationary series	–	Block bootstrap
Kaufman et al. [10]	2012	– (framework)	–	Leakage formulation and detection
This work	2026	nycflights13 (2013)	20:00 on $D-1$	All of the above, applied to delays

a flight that leaves late but makes up time is not a late arrival.

- **Cancelled and diverted flights.** They have no arrival delay, so they are dropped from the target; but they stay in the *schedule* counts, because a cancellation is not known the evening before. A sensitivity analysis counts them as delayed instead (Sec. IX).
- **Consequence for modelling.** A moderate imbalance (∼1

TABLE V
RELATED WORK, CALIBRATION, SHIFT, EVALUATION AND LEAKAGE PAPERS (2/2): METRICS, CALIBRATION, TEMPORAL HANDLING AND LIMITATION.

Paper	Metrics	Calibration reported	Leakage / temporal handling	Limitation (for our purpose)
Brier	Mean squared probability error	Yes (score)	–	Aggregate score only
Platt	Probabilistic outputs	Yes	i.i.d. calibration set	Static fit
Zadrozny & Elkan	Calibration quality	Yes	i.i.d.	Static fit
Niculescu-Mizil & Caruana	Calibration, ranking	Yes	i.i.d. splits	Static calibration
Kull et al.	Calibration quality	Yes	i.i.d.	Static; not temporal
Guo et al.	ECE, NLL	Yes (ECE)	i.i.d. test sets	No temporal shift
Ovadia et al.	Accuracy, NLL, ECE	Yes	Explicit dataset shift	Deep nets; not delay data
Rabanser et al.	Detection power	–	Explicit shift	Detection, not calibration
Gama et al.	–	–	Drift handling	Survey; no calibration focus
Gneiting & Raftery	–	Theory	–	Theory only
Davis & Goadrich	PR, ROC	–	–	Theory only
Saito & Rehmsmeier	PR vs ROC	–	–	Not a delay study
Tashman	–	–	Rolling origin	Forecasting, not classification
Bergmeir & Benítez	Forecast error	–	Blocked CV	Time-series regression
Künsch	–	–	Dependent data	Statistical theory
Kaufman et al.	–	–	Leakage theory	General; not flight-specific
This work	PR-AUC, ROC-AUC, Brier, ECE, lift	Yes, static and rolling	Decision clock + leakage tests	Single year, no weather

in 4): PR-AUC is the main ranking metric, and class weights are not used so that predicted probabilities stay interpretable.

C. Patterns that drive the design

- **Time of day.** The delay rate rises from 10.5% at 05:00 to 37.4% at 21:00 (Fig. 1) because delay propagates through the day. ⇒ cyclic time-of-day features plus the raw hour.
- **Season.** Monthly rates range from 13.7% (September) to 33.5% (December). September–October are calm and December is extreme, so a chronological split is a built-in stress test for calibration; with a single year, month effects cannot be separated from one-off events.
- **Carrier.** Rates range from 12.9% (HA) to 38.8% (F9) and some carriers are tiny (OO: 29 flights). ⇒ one-hot encoding with a minimum frequency, and shrinkage of history rates.
- **Memory.** The lag-1 autocorrelation of the daily delay rate is 0.40 and falls to about 0.12 over lags 8–14: disruption is bursty with short memory. ⇒ day-level history features alongside entity-level ones.
- **Congestion and the time-of-day confound.** Busier banks have more delays, but busy banks cluster in the evening where delays are high anyway. Splitting banks into terciles *within each departure hour* holds the hour fixed (Fig. 2): the busiest third is still 4.5 percentage points later than the quietest, and later in 13 of 16 hours. Part of the raw effect is time of day, but not all of it.
- **Multicollinearity.** The history rates are strongly inter-correlated (nested windows on the same key; network and origin rates that move together), some at $|r|$ close to 1 (Fig. 3). ⇒ logistic regression must be regularised, and single-feature importance is misleading, so we use *grouped* permutation importance.

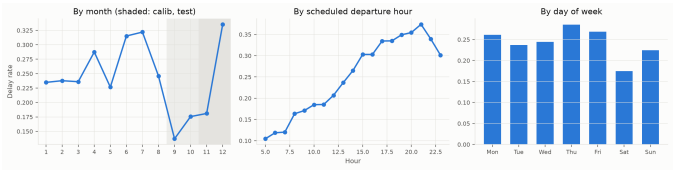


Fig. 1. Delay rate by month, hour and weekday.

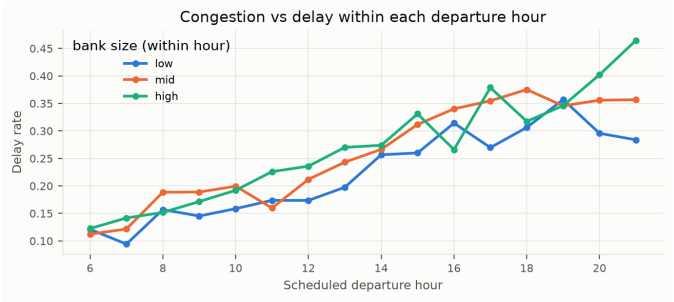


Fig. 2. Delay rate by scheduled-bank size, holding the departure hour fixed.

IV. DECISION CLOCK AND LEAKAGE

Leakage is information about the target that should not be available when the prediction is made [10]. In delay data there are two kinds: *column* leakage (post-departure columns such as the actual departure delay) and *temporal* leakage (past outcomes used before they were known). The first is removed by construction; the second needs an explicit clock.

A. Definition

- **Decision time.** 20:00 NYC on $D-1$ for every flight of day D.
- **Known when.** An arrival delay is known at the *actual gate arrival*, converted to the NYC clock; a cancellation is

TABLE VI
FROM EXPLORATORY OBSERVATION TO DESIGN DECISION.

Observation	Design decision
Missingness equals cancellations and diversions	Drop from the target, keep in schedule (congestion) counts, no imputation
Strong time-of-day gradient (10% to 37%)	Cyclic departure and arrival time plus the raw hour
One year; seasonal months; calm Sep–Oct, extreme December	No month feature; chronological split; calibration studied under shift
Short-memory daily shocks (lag-1 autocorrelation ≈0.4)	One-day and seven-day lagged rates (day-level history group)
Stable carrier, route and aircraft propensities; some tiny groups	7- and 14-day entity rates with Bayesian shrinkage (CV-tuned m, window); one-hot with minimum frequency 50
Congestion correlates with delay but is confounded with the hour	Scheduled-bank counts, kept and tested by ablation
Leakage risk from post-departure columns; 31% of $D-1$ arrivals land after 20:00	Decision clock at 20:00 on $D-1$; outcomes enter history only once known; two leakage tests
History rates strongly inter-correlated	L2-regularised logistic regression; grouped, not per-feature, permutation importance

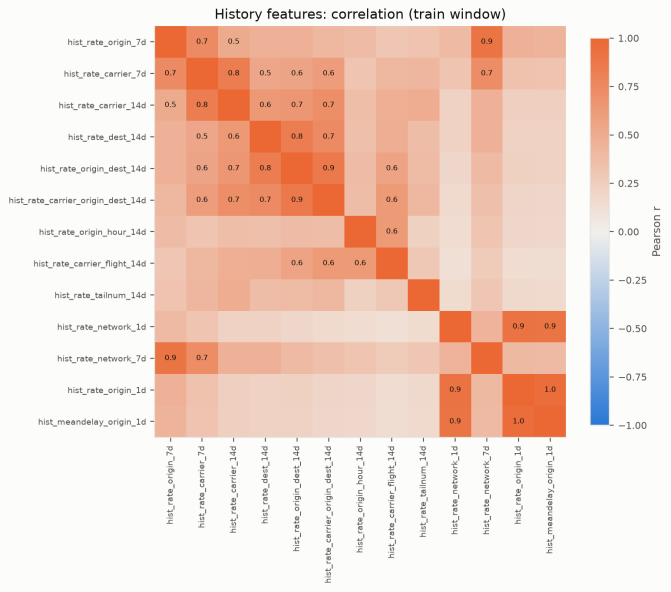


Fig. 3. Correlation of the history features (training window only).

known at the scheduled departure (or the actual departure for flights that left).

- **Availability date.** The first target day whose decision time is at or after the known-time t:

$$a(t) = \lceil t - 20\,\mathrm{h} \rceil_{\mathrm{day}} + 1 \text{ day}.$$

- **Windows roll over availability dates, not flight dates.** A flight dated $D-1$ that lands at 22:30 has $a(t) = D+1$ and cannot help predict D. The timetable of day D (including flights that will be cancelled) is allowed; anything that happens on D is not.

B. Time zones and daylight saving

- **Problem.** `arr_time` is in the *destination's* local time and the dataset has no time-zone table.
- **Solution.** A **fixed lookup table** of 105 destination offsets relative to NYC (Table VII) plus the 2013 daylight-saving rule (10 March 02:00 to 3 November 02:00; HNL, PHX, SJU, BQN, PSE and STT do not observe DST). The arrival on the NYC clock is scheduled departure plus departure delay plus the local-time difference corrected for the offset.
- **Why not estimate offsets from the data.** Our first version did, and past features then depended on future data; the future-dependence test caught it. The table is static geography. It reproduces the data-derived offsets (checked in the notebook and by a test), and the arrival times it gives imply plausible taxi times for flights with an air time.

TABLE VII
DESTINATION OFFSETS RELATIVE TO NYC IN THE LOOKUP TABLE.

Offset (min)	Destinations
0 (Eastern)	60
−60 (Central)	21
−120 (Mountain)	8
−180 (Pacific, incl. PHX)	14
−240 (ANC)	1
−360 (HNL)	1

C. Size of the leak

- **Which outcomes are late.** 31.1% of $D-1$ arrivals land after 20:00 (usable only from $D+1$) and are delayed 38.1% of the time, against 18.3% for arrivals known by 20:00. The unknown outcomes are the *bad evening* the model should anticipate.
- **Measured, not argued.** An earlier version used all flights dated $D-1$ for day D. With the current code we move the clock (Table XVIII, last rows): a 24:00 clock (every $D-1$ arrival counts) gives rolling-CV PR-AUC **0.4812** for logistic regression against 0.4705 at 20:00, a leak worth about **0.011**.
- **Why not simply lag two days.** Discarding all of $D-1$ (a 00:00 clock) gives 0.4587, lower by 0.012 and in all four folds, and throws away the 69% of $D-1$ outcomes that *are* known by 20:00. The clock keeps the usable information and drops only the unusable.

TABLE VIII
FEATURE GROUPS (n = NUMBER OF FEATURES).

Group	n	Content
Identifiers	3	carrier, origin, destination
Schedule	10	cyclic time-of-day and weekday, departure hour, weekend, days to holiday (cap 14), log distance
Congestion	7	scheduled flights in the origin-hour, ± 1 h bank and day, destination and carrier volumes, leg index
Entity history	18	smoothed delay rate and log-volume by route, carrier-route, carrier, destination, origin$\times$hour, flight number, tail number
Day-level history	7	network and per-origin rate over the last known 24 h, 7-day network rate, cancellation rates, mean delay

D. Leakage tests

- **(a) Future outcomes.** Overwrite every post-departure column from 15 June onward with random values and rebuild all features. Changed feature cells up to and including the cutoff: **0** (bit-for-bit identical); after the cutoff: 4,558,344, so the test is not vacuous.
- **(b) Decision clock.** For a target day D, randomise the arrival outcomes of $D-1$ flights that landed after 20:00. Changed cells on day D for 14 Feb, 11 Jul and 20 Dec: **0**.
- **Positive control.** Randomising the $D-1$ flights that landed *before* 20:00 changes 12,428, 13,078 and 12,740 cells; a test that cannot fail proves nothing, so this control confirms the check is sensitive.

V. FEATURES

A. Feature groups

There are 45 features in five groups (Table VIII). Groups matter beyond bookkeeping: ablation and permutation importance operate on whole groups, because correlated members of one group can cover for each other.

- **Schedule.** Time of day and weekday are encoded as sine/cosine pairs so that 23:59 and 00:01 are neighbours; the raw hour is kept as well. Days to the nearest US federal holiday are capped at 14 days.
- **Congestion.** Counts of *scheduled* flights in the same origin-hour and neighbouring banks; cancellations are not subtracted, because they are unknown the evening before. The leg index (the aircraft's nth departure of the day) is a partial proxy for rotation.
- **Entity history.** Smoothed late rates and log-volumes for entities that carry persistent delay propensity: routes, carriers, destinations, origin-hours, flight numbers and tail numbers.
- **Day-level history.** Network-wide and per-origin rates over the last *known* 24 hours, plus a 7-day rate: the day's disruption state as far as the clock allows.

B. Shrinkage of entity rates

- **The problem.** A tail number seen once and late has an empirical rate of 100%; a raw rate from a handful of flights is extremely noisy.
- **Beta-binomial shrinkage.** Treat the entity's late probability as having a Beta prior centred on the network rate π_D with strength m pseudo-flights; the posterior mean is

$$\hat{r}_{k,D} = \frac{\text{late}_k + m\,\pi_D}{\text{flights}_k + m},$$

where the counts cover outcomes *available* for D within the window and π_D is the network delay rate over the 28 days before D (also past-only).

- **Behaviour.** With few flights the estimate stays near π_D; with many, it approaches the entity's own rate. m sets how many flights it takes to trust the entity.
- **Worked example.** EWR$\rightarrow$CAE is a thin route with one evening flight every three to four days; every flight lands after 20:00, so an outcome is usable only from two days later. With one prior late flight and $\pi_D = 0.247$, $m = 5$ gives $(1 + 5 \times 0.247)/6 = 0.372$, not 1.0.

C. Joint tuning of m, window and C

- **Why joint.** Hand-picked values ($m = 20$, 28 days) were replaced by a rolling-CV search. The best C depends on how noisy the features are, so each (m, w) pair is scored at its own best C of the logistic grid; fixing C could mislead the choice.
- **Search.** $m \in \{5, 20, 50\}$, window $w \in \{14, 28, 56\}$ days, nine pairs (Table XVIII).
- **Result.** Less shrinkage is better at every window, and the 14-day window is best at every m. The CV-best triple is $m = 5$, $w = 14$, $C = 0.1$ (0.4705). An earlier search fixed $C = 0.001$; re-tuning jointly selected the *same* pair, so **no decision changed**.
- **How large is the gain.** Small relative to the fold spread (the old hand-picked values scored 0.4657); but the rule is "CV-best", and the pipeline aborts if the configuration disagrees with the table, so the choice is mechanical.

D. Design choices

- **No month features.** Training covers months 1–8, so months 9–12 would be an extrapolation, and each month is one realisation. CV agrees (Table XIX): adding cyclic month features lowers logistic regression from 0.4705 to 0.4686 (worse in two of four folds) and is a wash for XGBoost (0.4450 to 0.4471). Seasonality reaches the model through the trailing history rates. An earlier test-based justification for this choice was withdrawn (Sec. VII).
- **Day-level history is its own group.** `hist_rate_network_1d` takes one value per day, shared by hundreds of flights, so its effective sample size is about the number of training days and a flexible model could use it as a date identifier. Keeping it separate lets the ablation test it in isolation.
- **Encoding.** One-hot identifiers with minimum category frequency 50; numeric features standardised (for logistic regression).

VI. Models and Training

A. Logistic regression

- **Model.** $P(y=1 \mid x) = \sigma(w^\top x + b)$, trained by minimising the L2-regularised log-loss, a convex problem with a unique minimum:

$$\mathcal{L}(w) = -\tfrac{1}{n}\textstyle\sum_i \left[y_i \log p_i + (1-y_i)\log(1-p_i)\right] + \tfrac{1}{2Cn}\|w\|^2.$$

- **Why regularise.** The history features are strongly correlated, which makes unregularised coefficients unstable. The penalty is equivalent to a zero-mean Gaussian prior on w (MAP estimation); smaller C shrinks harder.
- **Why it is a strong baseline here.** The history features are already non-linear summaries (smoothed rates), so a linear model on top can capture most of the signal, and it is fast, convex and interpretable (coefficients, grouped permutation importance).
- **Choosing C.** From $\{3{\times}10^{-4}, 10^{-3}, 3{\times}10^{-3}, 10^{-2}, 0.1, 1\}$ by rolling CV (Table XX, Fig. 12); the curve is flat above $C \approx 0.003$ and the fold spread is far larger than the difference between values, so $C = 0.1$ is the marginal maximum, not a sharp optimum.

B. XGBoost

- **Model.** An additive ensemble of trees $F_T(x) = \sum_t f_t(x)$ fitted stage-wise [1], [8]. At stage t it minimises a second-order Taylor approximation of the loss plus a complexity penalty:

$$\tilde{\mathcal{L}}^{(t)} = \textstyle\sum_i \left[g_i f_t(x_i) + \tfrac{1}{2} h_i f_t(x_i)^2\right] + \gamma T + \tfrac{1}{2}\lambda\|w\|^2,$$

with $g_i = p_i - y_i$ and $h_i = p_i(1-p_i)$ for the log-loss.

- **Leaf weights and split gain.** For a fixed tree structure the optimal leaf weight is $w_j^* = -G_j/(H_j+\lambda)$, where G_j, H_j sum gradients and Hessians in the leaf. A split is accepted when

$$\text{Gain} = \tfrac{1}{2}\left[\tfrac{G_L^2}{H_L+\lambda} + \tfrac{G_R^2}{H_R+\lambda} - \tfrac{(G_L+G_R)^2}{H_L+H_R+\lambda}\right] - \gamma > 0.$$

- **Role of the knobs.** *Depth* caps interaction order (depth 2 means pairwise interactions at most). *Minimum child weight* is a minimum Hessian mass per leaf, which guards against fitting tiny groups on a weak, noisy signal. *Column subsampling* and λ regularise; *learning rate* trades the number of trees against smoothness.
- **Search.** 48 combinations of depth $\{2, 3, 5\}$, minimum child weight $\{20, 100\}$, column subsample $\{0.5, 0.8\}$, $\lambda \in \{1, 10\}$ and learning rate $\{0.03, 0.1\}$ (row subsample fixed at 0.8); 16 sampled at random, each with up to 3,000 trees and early stopping (100 rounds) on each fold's validation month. The tree count of a configuration is the mean of its four best iterations.
- **Selected.** Depth 2, learning rate 0.1, minimum child weight 100, column subsample 0.8, $\lambda = 10$, **143 trees**; in-fold CV 0.4754 ± 0.0673. The heavily constrained choice says the signal is weak and smooth, and the top configurations differ by less than one fold standard deviation.
- **Why log-loss.** A strictly proper scoring rule [18]: its expectation is minimised only by the true probability, so training pushes toward calibration on the training distribution. The Brier score [17] is also proper; accuracy and AUC are not.

C. Reference models

- **Prior.** The training base rate for every flight (PR-AUC equals the evaluated window's base rate; ROC-AUC is 0.5). It shows what a model with no information achieves, including on calibration metrics.
- **Route history.** The smoothed 14-day carrier-route delay rate alone, with no model. A learned model must beat this simple, transparent heuristic to justify itself.
- **Placement.** Both appear **beside every learned model** in every table and figure.

D. Calibration methods

- **Platt scaling** [11] fits $\sigma(a \operatorname{logit} p + b)$ on held-out scores: two parameters, smooth and monotone. **Temperature scaling** is its one-parameter special case [2].
- **Isotonic regression** [12], [13] fits any monotone step function by pool-adjacent-violators: more flexible, but it needs more data and can overfit.
- **Beta calibration** [14] is a richer parametric family; we mention it as an alternative and do not use it.
- **Shared assumption.** All of them assume the calibration data resemble deployment data. Sec. IX shows what happens when that fails.

VII. Evaluation Protocol

A. Split and cross-validation

- **Chronological split** (Table XXI): train Jan 8–Aug (211,684 rows), calibration Sep–Oct (55,628), test Nov–Dec (53,991). The first week is excluded as warm-up for the history windows.
- **Base rates differ by design.** 26.5% in training, 15.7% in calibration, 25.8% in test (18.1% in November, 33.5% in December): the split is a deliberate stress test.
- **Rolling-origin CV** inside the training window [21], [22]: train Jan–Apr $\rightarrow$ validate May; Jan–May $\rightarrow$ Jun; Jan–Jun $\rightarrow$ Jul; Jan–Jul $\rightarrow$ Aug. Each fold validates on a month strictly after its training data.
- **Abandoned design.** A single validation month (September) was dropped: it is the calmest month and its best XGBoost tree count differed from October's on nearly flat curves (Fig. 13).

B. Metrics

- **PR-AUC** is average precision, $AP = \sum_k (R_k - R_{k-1}) P_k$, a step-wise sum, not a trapezoidal area; the no-skill value is the base rate [19], [20]. **ROC-AUC** is reported alongside; it ignores the base rate.
- **Brier score** $= \frac{1}{n}\sum_i (p_i - y_i)^2$ [17], a proper score combining calibration and sharpness.
- **ECE** [2] is $\sum_b \frac{|B_b|}{n} |\bar{y}_b - \bar{p}_b|$ over probability bins. It is a biased estimator (sampling noise always adds, growing with the number of bins) and binning-dependent, so we report

both equal-width (15 bins) and equal-mass versions. It is a *marginal* check: it rewards a constant prediction when base rates happen to agree.

- **Lift** = policy precision / base rate.
- **Oracle recall@p.** The best recall at precision $\geq p$ with the threshold tuned *on the evaluated labels*. It is a point on the PR curve for comparing models, never an operating result.

C. Operating policies (fixed before test)

- **Primary: daily top 10%.** Flag the riskiest 10% of each day's flights, as a desk with fixed review capacity would. It uses within-day ranks only, so it is immune to base-rate shift and to any monotone recalibration within a day. Exact ties are broken by a seeded random draw; an order-based tie-break would let a constant score pick the day's latest flights and appear skilful (a bug we found and fixed).
- **Secondary: frozen threshold.** A score threshold frozen on Sep–Oct to reach precision 0.40. A target of 0.50 was tried first and is unreachable there (the base rate is 15.7%).

D. Calibration rules

- **Static rule (fixed before test).** Fit Platt and isotonic on September, check ECE on October, keep raw scores unless the best recalibrator lowers October ECE by more than 0.005, then refit on Sep–Oct.
- **Rolling Platt (adaptive alternative).** For each day D, refit Platt on outcomes *available* by D's decision time over the trailing N days and apply it to D's flights. It uses the same availability dates as the features and is causal. $N = 28$ was fixed a priori; $N = 14$ and $N = 56$ are sensitivity only.

E. Uncertainty

- **Day-block bootstrap** (500 replicates) [23]. Flights on one day share weather, air-traffic programmes and cascading delays, so their errors are correlated; resampling flights would treat about 885 correlated flights per day as independent and give intervals far too narrow.
- **Paired differences.** Model differences are computed on the same resampled days, which removes the shared day-to-day variation.

F. Development protocol and disclosure

- **The test window was not touched only once.** Test metrics were visible in: (1) the first single-September design; (2) a month-feature comparison, the original and now withdrawn justification for omitting month; (3) feature-subset diagnostics that shaped the day-level hypothesis; (4) lowering the precision target from 0.50 to 0.40, decided because 0.50 is unreachable on Sep–Oct, yet test recall was printed; (5) before any calibration rule existed; (6) one pipeline run where the final-model comparison still used XGBoost's optimistic in-fold score, corrected for a methodological reason found from CV alone; (7) when adding the prior to the policy table exposed the order-based tie-break.
- **Evidence is therefore separated.** Everything used to *decide* (clock, m, window, C, XGBoost configuration, final model, calibration rule and its outcome, month removal, ablation claims) comes from **CV or Sep–Oct**. The rolling-calibration window ($N = 28$), the 10% share and the precision target were set a priori or on Sep–Oct. **Test results are exploratory** and small differences should be read with caution.

VIII. Evidence Used for Selection (CV and Sep–Oct)

A. Final-model choice

- **Scored as deployed.** XGBoost's search score (0.4754) early-stops on the validation fold itself, which tunes the tree count on the scored data and makes the score optimistic. With its fixed, fold-averaged tree count it is **0.4450**. Logistic regression, with no in-fold tuning, is **0.4705**. References: route history 0.3476, prior 0.2775 (Table IX).
- **Per-fold differences** (LogReg − XGBoost): $+0.031$, $+0.075$, -0.001, -0.003; mean $+0.026$; fold sd about 0.07. LogReg leads in two of four folds.
- **Conclusion.** The two models are **not statistically distinguished**. LogReg is chosen by the rule declared before test (higher CV mean as deployed, recorded in the configuration; the pipeline aborts if CV ever disagrees), not by demonstrated superiority.
- **A disclosed reversal.** With the optimistic in-fold score the decision would have reversed (XGBoost 0.4754 against 0.4705); that is why the correction is disclosed.

B. Calibration rule outcome

- **LogReg → Platt.** October ECE improves by 0.026 (well above the 0.005 margin). Raw logistic regression overpredicts in October because it carries the 26.5% training base rate into a calmer month.
- **XGBoost → raw.** The best recalibrator makes October ECE *worse* (-0.025), so the rule keeps raw scores.
- **Names.** "LogReg (Platt, rule-selected)" and "XGBoost (raw, rule-selected)" denote these variants throughout (Table X). Sep–Oct is also where the secondary policy's threshold was frozen.

C. Feature families

- **LogReg gains in CV.** Congestion $+0.021$ (4/4 folds), entity history $+0.020$ (4/4), day-level history $+0.011$ (3/4) (Fig. 4). This CV evidence, not a test difference, supports keeping every family and replaces the earlier test-based justification of the day-level group.
- **XGBoost.** Congestion $+0.012$ (3/4); entity history -0.002 (2/4); day-level -0.004 (3/4). No claim rests on its ablation.
- **Complementarity.** History alone (logistic regression, 25 features) reaches 0.4478, below schedule plus history: the families complement each other.
- **Grouped permutation importance** (Sep–Oct; Table XXIII, Fig. 5) shuffles a whole family with one permutation so correlated members cannot cover for each other. For the final model: congestion 0.046, schedule

TABLE IX
ROLLING-CV PR-AUC (VALIDATION MONTHS MAY–AUG), MODELS SCORED AS DEPLOYED.

Model	Calibration	CV PR-AUC	CV sd	May	Jun	Jul	Aug
Prior (base rate)	none	0.277	0.048	0.227	0.315	0.322	0.246
Route history	none	0.348	0.060	0.277	0.391	0.403	0.319
LogReg (as deployed)	raw	0.471	0.071	0.435	0.503	0.552	0.392
XGBoost (as deployed, fixed trees)	raw	0.445	0.073	0.404	0.428	0.552	0.395
XGBoost (in-fold early stopping, optimistic; not used)	raw	0.475	0.067	0.449	0.496	0.557	0.399

TABLE X
CALIBRATION RULE: FIT ON SEPTEMBER, EVALUATED ON OCTOBER.

Model	Calibrator	Oct ECE	Oct ECE (eq.-mass)	Oct Brier	ECE gain vs raw
logreg	raw	0.061	0.061	0.140	0.000
logreg	platt	0.035	0.035	0.138	0.026
logreg	isotonic	0.036	0.036	0.139	0.025
xgboost	raw	0.022	0.025	0.139	0.000
xgboost	platt	0.046	0.046	0.141	-0.024
xgboost	isotonic	0.047	0.047	0.141	-0.025

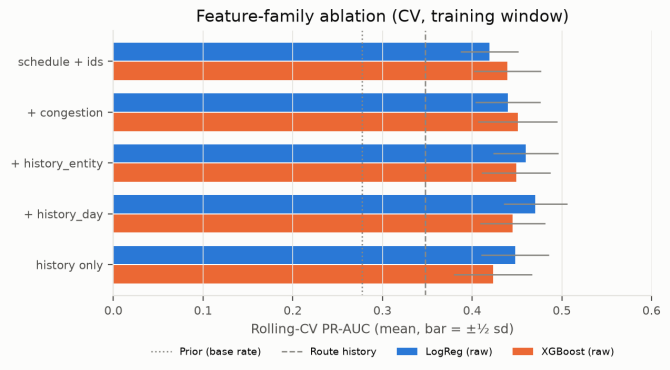


Fig. 4. Rolling-CV PR-AUC of cumulative feature sets, with prior (dotted) and route-history (dashed) references.

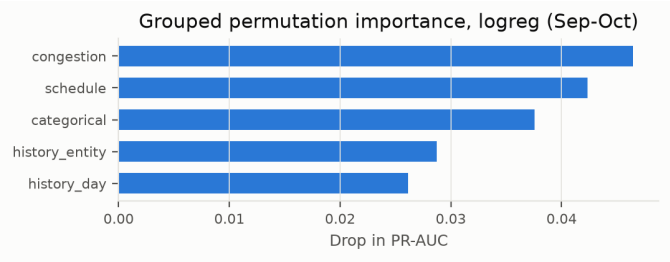


Fig. 5. Grouped permutation importance of the final model.

0.042, identifiers 0.038, entity history 0.029, day-level history 0.026. For XGBoost, schedule dominates (0.054). Descriptive only; not a selection criterion.

IX. EXPLORATORY TEST-SET FINDINGS (NOV–DEC)

Nothing in this section chose anything: the model, the calibration rule, the policies and N were fixed beforehand. Treat every number as exploratory.

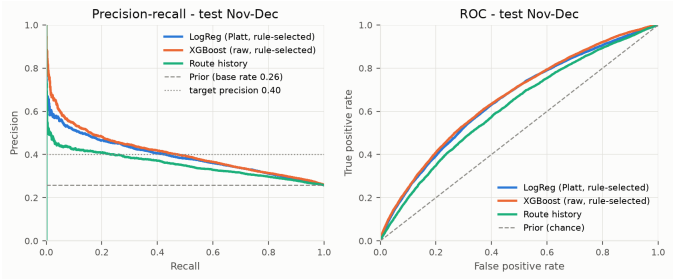


Fig. 6. Precision–recall and ROC curves on test. Dashed grey: prior.

A. Headline metrics

- **Results.** The final model, LogReg (Platt, rule-selected): PR-AUC **0.394**, ROC-AUC 0.654. XGBoost (raw, rule-selected): 0.407 and 0.662. Route history: 0.353 and 0.621. Prior: 0.258 and 0.500 (Table XI, Fig. 6). Both learned models beat both references.
- **Platt is monotone.** LogReg (Platt) and LogReg (raw) rank identically; they differ in calibration (ECE 0.082 against 0.019), which is the subject of the calibration subsection below.
- **The prior's low ECE (0.007) is not skill.** A constant at the training base rate is calibrated when the test base rate (25.8%) happens to be close to it; its equal-mass ECE (0.084) and its PR-AUC expose it.
- **Day-block bootstrap** (Table XII). XGBoost minus LogReg is $+0.013$ PR-AUC with 95% CI $[-0.010, +0.041]$, which includes zero, in line with CV; for ROC-AUC the interval is $[-0.012, +0.029]$.
- **By month** (Table XXII, appendix). The test window is split into November and December, because the base rate differs strongly (18.1% vs 33.5%).

TABLE XI
TEST METRICS (NOV–DEC). ORACLE REC. IS THE BEST RECALL AT PRECISION ≥ 0.40 WITH THE THRESHOLD TUNED ON TEST LABELS: COMPARISON ONLY.

Model	Calibration	PR-AUC	ROC-AUC	Brier	ECE	ECE (eq.-mass)	Oracle rec.@0.40
Prior (base rate)	none	0.258	0.500	0.192	0.007	0.084	0.000
Route history	none (smoothed rate)	0.353	0.621	0.187	0.037	0.038	0.244
LogReg (raw)	raw	0.394	0.654	0.181	0.019	0.022	0.430
LogReg (Platt, rule-selected)	Platt (rule-selected)	0.394	0.654	0.187	0.082	0.082	0.430
LogReg (rolling Platt 28d)	rolling Platt 28d (causal)	0.392	0.657	0.181	0.027	0.026	0.432
XGBoost (raw, rule-selected)	raw (rule-selected)	0.407	0.662	0.180	0.027	0.027	0.474
XGBoost (rolling Platt 28d)	rolling Platt 28d (causal)	0.396	0.659	0.180	0.024	0.023	0.416

TABLE XII
DAY-BLOCK BOOTSTRAP ON TEST (500 REPLICATES).

Metric	XGBoost 95% CI	LogReg 95% CI	XGB−LR 95% CI	P(XGB not better)
PR-AUC	[0.348, 0.465]	[0.34, 0.45]	[-0.01, 0.041]	0.156
ROC-AUC	[0.633, 0.691]	[0.62, 0.682]	[-0.012, 0.029]	0.208

B. Operating policies

- **Primary (daily top 10%), LogReg.** Precision **0.444**, recall 0.173, lift **1.72**. XGBoost: 1.68; route history 1.39; prior 0.94, i.e. random (Table XIII).
- **Stable relative precision.** Relative precision is stable between Sep–Oct (precision 0.303, lift 1.93) and Nov–Dec; absolute precision rises with the base rate.
- **Secondary (frozen threshold).** LogReg flags 3.8% of flights in Sep–Oct but 7.5% in Nov–Dec, because the score distribution moved with the base rate: a fixed threshold **does not transfer** across seasons. The prior never reaches precision 0.40, so its threshold is infinite and it flags nothing.

C. Calibration under seasonal shift

- **Static Platt.** Fitted on calm Sep–Oct (15.7% late), it scales risk *downward*: right for November (ECE 0.021) and badly wrong for December (**0.142**; base rate 33.5%). Isotonic behaves like Platt. **The static rule fails in December.**
- **Raw LogReg** looks well calibrated overall (0.019) only because the training and test base rates happen to be close: a coincidence of this split, not a robust property.
- **Rolling Platt** ($N = 28$) tracks the shift: test ECE 0.027 for LogReg (November 0.007, December 0.050) and 0.024 for XGBoost. $N = 14$ adapts faster (0.021) and $N = 56$ lags (0.051), but only $N = 28$ was declared in advance.
- **Shape versus level.** Raw predictions follow the *shape* of the season but compress it toward the middle (Fig. 8).
- **Interpretation.** This matches the known behaviour of post-hoc calibration under dataset shift [2], [3]: a frozen calibrator breaks when the calibration data stop resembling deployment data, and the remedy is to recalibrate continuously on recent *known* outcomes (Table XIV, Fig. 7).

D. Ablation on test

- **LogReg.** Entity history is confirmed ($+0.032$, CI $[+0.015, +0.049]$); congestion ($+0.007$) and day-level history ($+0.009$) have intervals that include zero. These are confirmations only; the decisions came from CV.
- **XGBoost.** Entity history shows $+0.052$ on test although CV showed no gain: an unresolved disagreement, so we make no claim for XGBoost's ablation (Table XV).

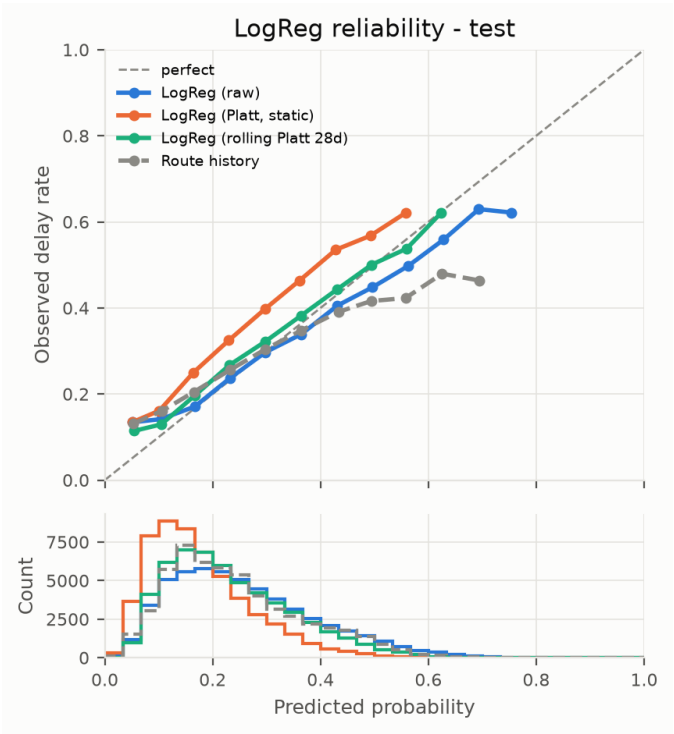


Fig. 7. Reliability of logistic regression on test: raw, static Platt, rolling Platt, route history.

E. Target definition

- **Cancelled and diverted counted as delayed.** The test base rate rises from 25.8% to 27.7% and PR-AUC with it (LogReg 0.446, XGBoost 0.455; the prior moves in step).

TABLE XIII
OPERATING POLICIES ON THE TEST WINDOW.

Model	Policy	Flag rate	Precision	Recall	Lift	Threshold
Prior (base rate)	daily top 10%	0.101	0.243	0.095	0.941	–
Prior (base rate)	frozen thr. (P≥0.4)	0.000	–	0.000	–	∞
Route history	daily top 10%	0.101	0.358	0.139	1.387	–
Route history	frozen thr. (P≥0.4)	0.059	0.425	0.097	1.646	0.469
LogReg (Platt, rule-selected)	daily top 10%	0.101	0.444	0.173	1.720	–
LogReg (Platt, rule-selected)	frozen thr. (P≥0.4)	0.075	0.493	0.143	1.910	0.331
XGBoost (raw, rule-selected)	daily top 10%	0.101	0.435	0.169	1.683	–
XGBoost (raw, rule-selected)	frozen thr. (P≥0.4)	0.062	0.521	0.125	2.019	0.446

TABLE XIV
CALIBRATION OF EVERY VARIANT ON TEST, NOVEMBER AND DECEMBER.

Model	ECE test	ECE Nov	ECE Dec	Brier test
Prior (base rate)	0.007	0.084	0.071	0.192
Route history	0.037	0.035	0.060	0.187
LogReg (raw)	0.019	0.055	0.060	0.181
LogReg (Platt, static)	0.082	0.021	0.142	0.187
LogReg (isotonic, static)	0.082	0.023	0.142	0.187
LogReg (rolling Platt 14d)	0.021	0.011	0.039	0.181
LogReg (rolling Platt 28d)	0.027	0.007	0.050	0.181
LogReg (rolling Platt 56d)	0.051	0.014	0.089	0.182
XGBoost (raw)	0.027	0.019	0.064	0.180
XGBoost (Platt, static)	0.081	0.033	0.131	0.186
XGBoost (isotonic, static)	0.079	0.034	0.126	0.186
XGBoost (rolling Platt 14d)	0.023	0.012	0.043	0.180
XGBoost (rolling Platt 28d)	0.024	0.013	0.038	0.180
XGBoost (rolling Platt 56d)	0.040	0.022	0.058	0.180

TABLE XV
ABLATION STEPS: CV GAIN (SELECTION EVIDENCE) AND TEST DIFFERENCE WITH 95% CI (EXPLORATORY).

Model	Calibration	Step	CV gain	Folds up	Test diff.	CI low	CI high
LogReg	raw	+ congestion	0.021	4/4	0.007	-0.009	0.027
LogReg	raw	+ history_entity	0.020	4/4	0.032	0.015	0.049
LogReg	raw	+ history_day	0.011	3/4	0.009	-0.009	0.027
XGBoost	raw	+ congestion	0.012	3/4	0.002	-0.010	0.015
XGBoost	raw	+ history_entity	-0.002	2/4	0.052	0.026	0.077
XGBoost	raw	+ history_day	-0.004	3/4	0.010	-0.024	0.045

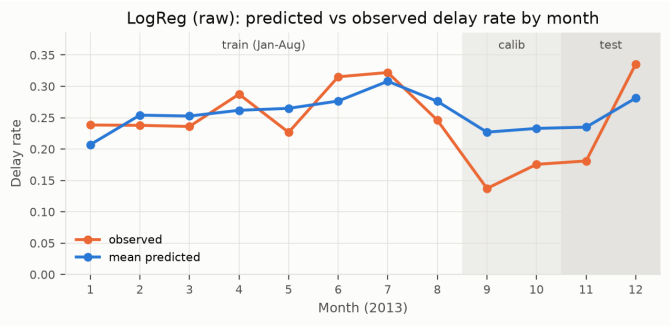


Fig. 8. Predicted vs observed delay rate by month (raw logistic regression).

- **Conclusions hold.** The XGBoost minus LogReg CI $[-0.013, +0.037]$ includes zero; the daily top-10% lift is 1.73 (LogReg), 1.71 (XGBoost) and 0.95 (prior); both learned models still beat both references (Table XVI).

F. *Covariate shift*

- **Methods.** Population stability index (PSI; training as reference; above 0.25 is a large shift) and adversarial validation (a classifier trained to separate training rows from later rows) [15] explain the calibration behaviour.
- **Adversarial validation** (Table XXIV). All features together separate training from later periods perfectly (AUC 1.00). By family: identifiers nearly stable (0.56), schedule moderately (0.78 against Sep–Oct, 0.92 against test), congestion strongly (0.98, 0.96) because daily volumes differ by season, and history features completely, as designed.
- **PSI** (Table XXV, Fig. 9). `days_to_holiday` shifts most into Nov–Dec (4.8) because Thanksgiving, Christmas and New Year fall in the test window; the history rates move most into Sep–Oct.
- **Implication.** Day-level history takes one value per day, so it identifies the period trivially: a memorisation risk for flexible models, and a reason a frozen calibrator breaks.

TABLE XVI
TARGET SENSITIVITY: CANCELLED AND DIVERTED FLIGHTS COUNTED AS DELAYED.

Target	Model	Base rate	PR-AUC	ROC-AUC	Top-10% lift
arr_delay≥15 (main)	Prior (base rate)	0.258	0.258	0.500	0.941
arr_delay≥15 (main)	Route history	0.258	0.353	0.621	1.387
arr_delay≥15 (main)	LogReg (raw)	0.258	0.394	0.654	1.720
arr_delay≥15 (main)	XGBoost (raw)	0.258	0.407	0.662	1.683
arr_delay≥15 OR cancelled/diverted	Prior (base rate)	0.277	0.277	0.500	0.946
arr_delay≥15 OR cancelled/diverted	Route history	0.277	0.374	0.621	1.359
arr_delay≥15 OR cancelled/diverted	LogReg (raw)	0.277	0.446	0.661	1.726
arr_delay≥15 OR cancelled/diverted	XGBoost (raw)	0.277	0.455	0.669	1.715

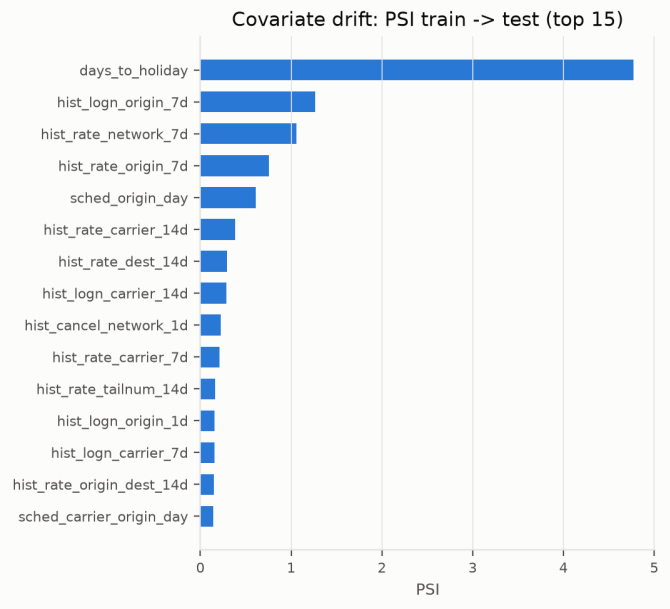


Fig. 9. PSI, training against the test window (top 15 features).

X. DECISION-SUPPORT ANALYSIS: RESCHEDULING CANDIDATES

The brief asks for rescheduling. The data contain no cost, aircraft-rotation or schedule-change model, so what we can demonstrate is whether the model's *daily ranking* concentrates delayed flights at the top, which is the quantity a capacity-limited review desk uses.

A. Design

- **Setup.** Each test day, the final model's ranking selects the top-$k\%$ flights as **reschedule / add-buffer candidates**, $k \in \{5, 10, 15, 20\}$. The test window has 61 days and about 885 flights per day, so $k = 5\%$ is about 45 flights and $k = 20\%$ about 177 flights per day.
- **Compared with.** The prior (random order) and route-history rankings, plus XGBoost.
- **Reported.** Flagged per day, precision, recall, lift, and delayed flights per 100 flagged (Table XVII).
- **Fixed in advance.** The k grid and the policy were specified before the analysis ran; there is no tuning on test.

B. Results

- **Top 5%** (≈45 flights/day). LogReg precision **0.469**, lift 1.82, i.e. **47 delayed per 100 flagged**, against 39 for route history and 25 for random.
- **Top 10%.** Precision 0.444, recall 0.173, lift 1.72 (identical to the primary policy of Sec. IX); about 40 delayed flights caught per day against about 32 for route history and 22 for random.
- **Top 20%** (≈177 flights/day). Precision 0.406, lift 1.57, recall 0.315, about 72 delayed flights caught per day (route history about 60, random about 45).
- **The trade-off.** Widening the desk from 5% to 20% catches more delays but with lower precision: the extra ≈133 flights per day are late about 38% of the time, against 47% for the top 5%. The ranking therefore degrades smoothly, with no cliff.
- **Ordering holds at every k.** LogReg ≥ XGBoost > route history > prior (Table XVII); LogReg versus XGBoost is not a distinguished difference (Sec. IX). Random flagging yields about 25 per 100 at every k, i.e. the base rate, as it should.

C. How to read it

- **Capacity as the design variable.** A desk that can review only 5% of flights should expect about 47 true delays per 100 reviews; one that can review 20% should expect about 41. The marginal value of added capacity falls, which is useful when staffing the desk.
- **Why rank-based.** Because it uses within-day ranks, the policy is unaffected by the December calibration failure (Sec. IX); a fixed probability threshold is not portable across seasons, a daily capacity share is.
- **Failure modes.** The ranking cannot know which delays are *preventable*; flights delayed by weather or air-traffic programmes are flagged but cannot be saved by rescheduling. Recall at 10% is only 0.173, so most delayed flights are never flagged.

D. What this shows — and does not

- **Shows: ranking quality** for a capacity-limited review desk.
- **Does not show: realised operational savings**. There is no cost model for delay or for a schedule change, so the benefit of moving or buffering a flight cannot be estimated.

TABLE XVII
RESCHEDULING-CANDIDATE ANALYSIS ON THE TEST WINDOW: DAILY TOP-$k\%$ BY RISK.

Model	Top share	Flagged/day	Precision	Recall	Lift	Delayed per 100
Prior (base rate)	5%	44.705	0.251	0.049	0.971	25.083
Prior (base rate)	10%	88.984	0.243	0.095	0.941	24.300
Prior (base rate)	15%	133.148	0.250	0.145	0.967	24.982
Prior (base rate)	20%	177.393	0.252	0.195	0.974	25.164
Route history	5%	44.705	0.386	0.076	1.495	38.614
Route history	10%	88.984	0.358	0.139	1.387	35.814
Route history	15%	133.148	0.346	0.201	1.339	34.597
Route history	20%	177.393	0.339	0.263	1.313	33.916
LogReg (Platt, rule-selected)	5%	44.705	0.469	0.092	1.817	46.938
LogReg (Platt, rule-selected)	10%	88.984	0.444	0.173	1.720	44.436
LogReg (Platt, rule-selected)	15%	133.148	0.423	0.246	1.636	42.256
LogReg (Platt, rule-selected)	20%	177.393	0.406	0.315	1.572	40.606
XGBoost (raw, rule-selected)	5%	44.705	0.464	0.091	1.797	46.425
XGBoost (raw, rule-selected)	10%	88.984	0.435	0.169	1.683	43.460
XGBoost (raw, rule-selected)	15%	133.148	0.416	0.242	1.611	41.603
XGBoost (raw, rule-selected)	20%	177.393	0.400	0.310	1.548	39.996

- **Missing for a full rescheduling study:** a cost model for delays and for schedule changes, feasibility constraints (gates, crews, slots, rotations), and the causal effect of an intervention on the flight's outcome. The project brief's exact wording of rescheduling has not been supplied; if it requires more than this, those parts are missing and are not invented here.

XI. DISCUSSION AND LIMITATIONS

A. What the evidence supports

- **A leakage-free ranking is worth acting on.** The final model flags the riskiest tenth of each day's flights with $1.72\times$ the base-rate precision, against $1.39\times$ for route history alone; the gap to the simple heuristic is the value added by the model.
- **Hand-built summaries leave little interaction structure for trees.** XGBoost is not distinguished from logistic regression in CV or on test: the smoothed history features already carry most of the usable signal. That is the gap a later deep temporal model would target.
- **Each feature family earns its place for LogReg.** Congestion, entity history and day-level history each improve it in CV, and entity history is also confirmed on test.
- **Calibration needs maintenance.** A frozen recalibrator fitted on a calm period breaks under seasonal prior shift; a rank-based daily policy is portable and a fixed threshold is not.

B. Negative and unresolved results

- **The static calibration rule fails** in December (ECE 0.142). It is reported as is.
- **XGBoost's ablation** disagrees between CV and test for entity history.
- **The final-model choice** would have reversed under the optimistic in-fold score.
- **The original month-feature and day-level justifications** relied on test numbers and were withdrawn and replaced by CV evidence.

C. Threats to validity

- **Internal.** The test window was seen during development (Sec. VII); we limit this by deciding only from CV and Sep–Oct, but the separation cannot be perfect. Four CV folds cannot separate models differing by thousandths.
- **External.** One year and three origins: each season and airport effect is observed once, and December is a single draw. Cross-year and cross-airport generalisation cannot be established.
- **Construct.** The label is a 15-minute threshold; the sensitivity analysis shows conclusions survive counting cancellations as delays, but other thresholds (30, 60 minutes) were not modelled.
- **Statistical.** Day-block intervals account for within-day correlation but not for correlation across neighbouring days; with 61 test days they are wide.

D. Limitations and assumptions

- **No weather, by design.** Observed weather for D would leak at the clock; only archived forecasts issued before 20:00 could be added. Weather is the main missing driver of day-to-day delay, and this is a stated gap rather than a defect.
- **Partial rotation.** Only NYC departures are visible, about 1.3 per aircraft per day.
- **Assumptions.** Tail-number assignment is known by 20:00 on $D-1$; holidays follow the US federal calendar; a cancellation is known at the scheduled departure and an arrival at the actual gate arrival; destination time zones come from a fixed table.

E. Future work

- Add archived weather *forecasts* issued before the clock.
- A deep temporal model on the same clock, to test whether it extracts structure that tree ensembles and logistic regression miss.
- Multi-year data to test cross-year generalisation of the calibration findings.

- A cost model and feasibility constraints for a genuine rescheduling study.

XII. Conclusion

- **Clock first.** Features admit only outcomes known at 20:00 on $D-1$; the quantified leak (≈0.011 CV PR-AUC) shows why this matters, and two tests (with a positive control) guard it.
- **Models.** Logistic regression is the declared final model (CV PR-AUC 0.4705 against 0.4450 for XGBoost as deployed), but the two are not statistically distinguished and the choice rests on a pre-declared rule.
- **Policy.** The daily top 10% reaches precision 0.444 and lift 1.72, and the ranking degrades smoothly as review capacity grows; a rolling recalibration (ECE 0.027) succeeds where the static rule (ECE 0.082) fails in December.
- **Honesty.** Selection evidence (CV, Sep–Oct) is kept apart from exploratory test findings, and every test exposure is disclosed.
- **Next.** Archived weather forecasts are the natural additional signal; a deep temporal model belongs to the next phase.

Reproducibility

`bash scripts/reproduce.sh` deletes generated artefacts and rebuilds the pipeline, four executed notebooks (which assert their tables against the saved results), the report figures and tables, the test suite, the number check, and this PDF. All randomness is seeded. Table XXVII lists the final configuration.

Appendix A
Additional Tables and Figures

Supporting material: full ablation (Table XXVI), test metrics by month (Table XXII), grouped permutation importance (Table XXIII), XGBoost gain (Fig. 10), XGBoost reliability (Fig. 11), the C curve (Fig. 12) and why a single validation month was abandoned (Fig. 13).

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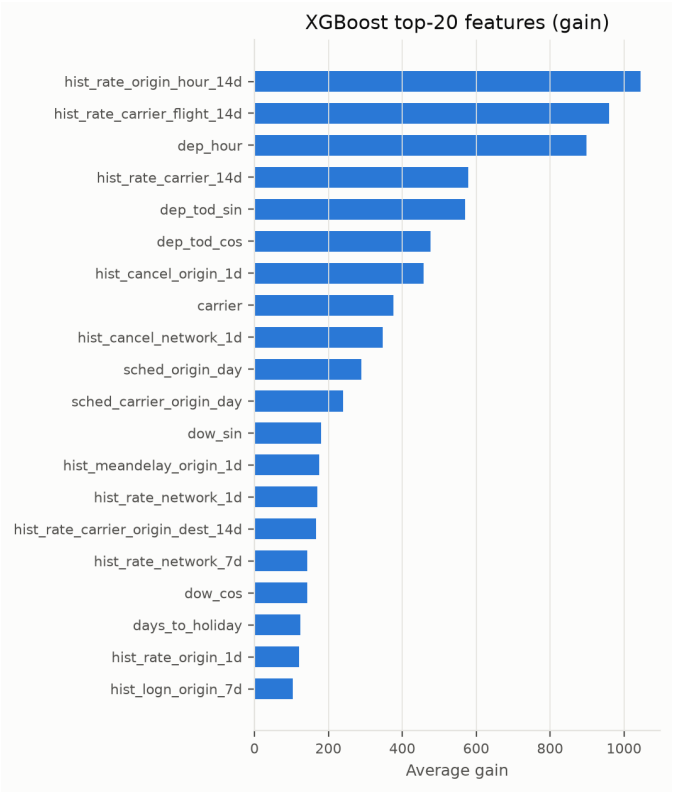


Fig. 10. XGBoost top-20 features by gain.

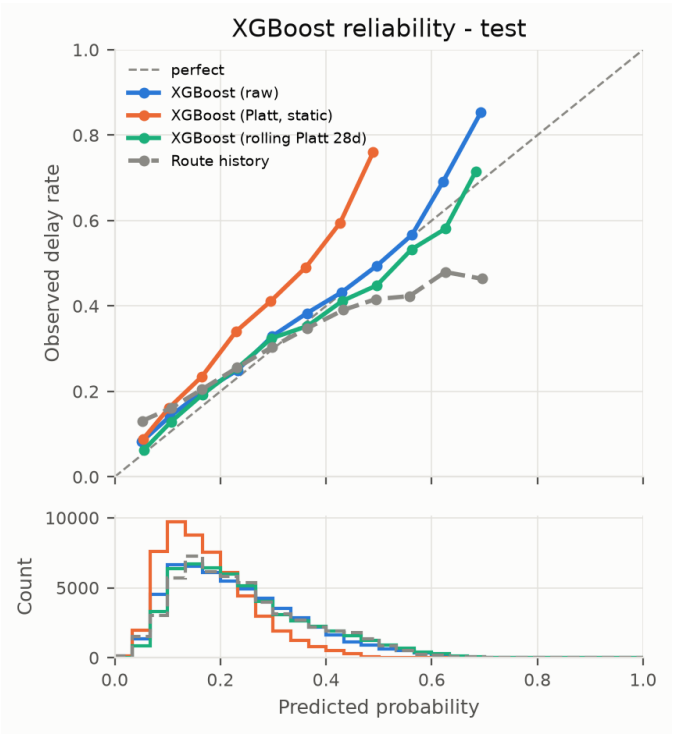


Fig. 11. Reliability of XGBoost on test: raw, static Platt, rolling Platt, route history.

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TABLE XVIII
HISTORY SETTINGS TUNED JOINTLY WITH C (ROLLING-CV PR-AUC OF LOGREG). CLOCK 20 IS DEPLOYED; THE LAST ROWS MOVE THE CLOCK TO 00:00 AND 24:00.

m	Window (d)	Clock (h)	Best C	CV PR-AUC	CV sd	CV (C=0.001)
5	14	20	0.100	0.471	0.071	0.468
5	28	20	0.001	0.467	0.064	0.467
5	56	20	0.010	0.468	0.069	0.466
20	14	20	0.010	0.469	0.071	0.467
20	28	20	0.001	0.466	0.064	0.466
20	56	20	0.010	0.467	0.069	0.465
50	14	20	0.010	0.465	0.072	0.463
50	28	20	0.001	0.463	0.064	0.463
50	56	20	0.010	0.464	0.069	0.463
5	14	0	0.003	0.459	0.073	0.458
5	14	24	0.010	0.481	0.072	0.480

TABLE XIX
ROLLING-CV PR-AUC WITHOUT AND WITH CYCLIC MONTH FEATURES.

Features	Model	May	Jun	Jul	Aug	CV PR-AUC
without month	logreg	0.435	0.503	0.552	0.392	0.471
without month	xgboost	0.404	0.428	0.552	0.395	0.445
with month	logreg	0.427	0.505	0.549	0.394	0.469
with month	xgboost	0.411	0.426	0.563	0.388	0.447

TABLE XX
LOGREG: ROLLING-CV PR-AUC BY C (MEAN, SD OVER FOUR FOLDS).

C	CV PR-AUC	CV sd
0.000	0.466	0.068
0.001	0.468	0.070
0.003	0.470	0.070
0.010	0.471	0.070
0.100	0.471	0.071
1.000	0.470	0.071

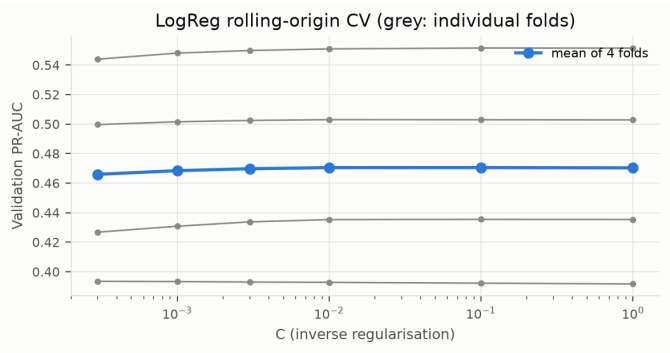


Fig. 12. Logistic regression rolling-CV PR-AUC against C.

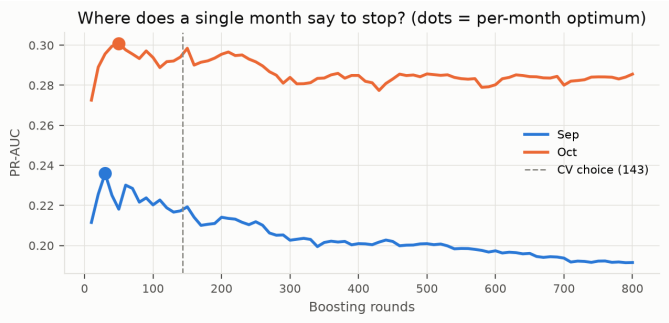


Fig. 13. Why a single validation month was abandoned: XGBoost PR-AUC on September and October vs number of trees.

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TABLE XXI
CHRONOLOGICAL SPLIT (CANCELLED AND DIVERTED FLIGHTS DROPPED).

Split	Flights	Delay rate	From	To
train	211684	0.265	2013-01-08	2013-08-31
calib	55628	0.157	2013-09-01	2013-10-31
test	53991	0.258	2013-11-01	2013-12-31

TABLE XXII
TEST METRICS BY MONTH.

Model	Split	PR-AUC	ROC-AUC	Brier	ECE
Prior (base rate)	Nov	0.181	0.500	0.155	0.084
Route history	Nov	0.221	0.560	0.149	0.035
LogReg (raw)	Nov	0.291	0.655	0.145	0.055
LogReg (Platt, rule-selected)	Nov	0.291	0.655	0.142	0.021
LogReg (rolling Platt 28d)	Nov	0.285	0.650	0.142	0.007
XGBoost (raw, rule-selected)	Nov	0.270	0.638	0.144	0.019
XGBoost (rolling Platt 28d)	Nov	0.261	0.631	0.144	0.013
Prior (base rate)	Dec	0.335	0.500	0.228	0.071
Route history	Dec	0.394	0.574	0.225	0.060
LogReg (raw)	Dec	0.456	0.632	0.217	0.060
LogReg (Platt, rule-selected)	Dec	0.456	0.632	0.233	0.142
LogReg (rolling Platt 28d)	Dec	0.431	0.603	0.220	0.050
XGBoost (raw, rule-selected)	Dec	0.468	0.635	0.216	0.064
XGBoost (rolling Platt 28d)	Dec	0.442	0.613	0.217	0.038

TABLE XXIII
GROUPED PERMUTATION IMPORTANCE: PR-AUC DROP ON SEP–OCT.

Group	LogReg	XGBoost
categorical	0.038	0.015
congestion	0.046	0.015
history_day	0.026	0.009
history_entity	0.029	0.018
schedule	0.042	0.054

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TABLE XXIV
ADVERSARIAL VALIDATION: AUC OF A CLASSIFIER SEPARATING TRAINING ROWS FROM LATER ROWS.

Comparison	Features	#feat.	Adversarial AUC
train vs calib	all features	45	1.000
train vs calib	categorical	3	0.562
train vs calib	schedule	10	0.785
train vs calib	congestion	7	0.983
train vs calib	history_entity	18	1.000
train vs calib	history_day	7	1.000
train vs test	all features	45	1.000
train vs test	categorical	3	0.562
train vs test	schedule	10	0.921
train vs test	congestion	7	0.963
train vs test	history_entity	18	0.998
train vs test	history_day	7	1.000

TABLE XXV
LARGEST POPULATION-STABILITY SHIFTS (TRAINING REFERENCE).

Feature	Group	PSI (Sep–Oct)	PSI (test)
days_to_holiday	schedule	0.261	4.769
hist_logn_origin_7d	history_entity	2.269	1.265
hist_rate_network_7d	history_day	5.147	1.058
hist_rate_origin_7d	history_entity	3.433	0.755
sched_origin_day	congestion	2.056	0.610
hist_rate_carrier_14d	history_entity	2.631	0.383
hist_rate_dest_14d	history_entity	2.143	0.292
hist_logn_carrier_14d	history_entity	0.472	0.289
hist_cancel_network_1d	history_day	1.229	0.222
hist_rate_carrier_7d	history_entity	1.699	0.209

TABLE XXVI
FULL ABLATION: CUMULATIVE FEATURE SETS FOR BOTH MODELS AND THE REFERENCES.

Model	Calibration	Features	#feat.	CV PR-AUC	CV sd	Test PR-AUC	Test ROC-AUC
Prior (base rate)	none	(baseline)	0	0.277	0.048	0.258	0.500
Route history	none (smoothed rate)	(baseline)	0	0.348	0.060	0.353	0.621
LogReg	raw	schedule + ids	13	0.419	0.064	0.345	0.613
LogReg	raw	+ congestion	20	0.440	0.073	0.352	0.615
LogReg	raw	+ history_entity	38	0.460	0.073	0.385	0.643
LogReg	raw	+ history_day	45	0.471	0.071	0.394	0.654
LogReg	raw	history only	25	0.448	0.075	0.390	0.657
XGBoost	raw	schedule + ids	13	0.439	0.075	0.343	0.619
XGBoost	raw	+ congestion	20	0.451	0.089	0.345	0.619
XGBoost	raw	+ history_entity	38	0.449	0.077	0.397	0.654
XGBoost	raw	+ history_day	45	0.445	0.073	0.407	0.662
XGBoost	raw	history only	25	0.423	0.088	0.380	0.646

TABLE XXVII
FINAL CONFIGURATION (ALL FIXED BY CV OR SEP–OCT, NONE BY TEST).

Item	Value
Decision clock	20:00 NYC on $D-1$
Target	arrival delay ≥ 15 min
Shrinkage m, window	5, 14 days
LogReg C	0.1
XGBoost	depth 2, lr 0.1, mcw 100, colsample 0.8, $\lambda = 10$, 143 trees
Final model	LogReg, Platt (rule-selected)
Rolling calibration	Platt, $N = 28$ days
Primary policy	daily top 10%
Seed	fixed (all randomness)